Dockerized Static Website

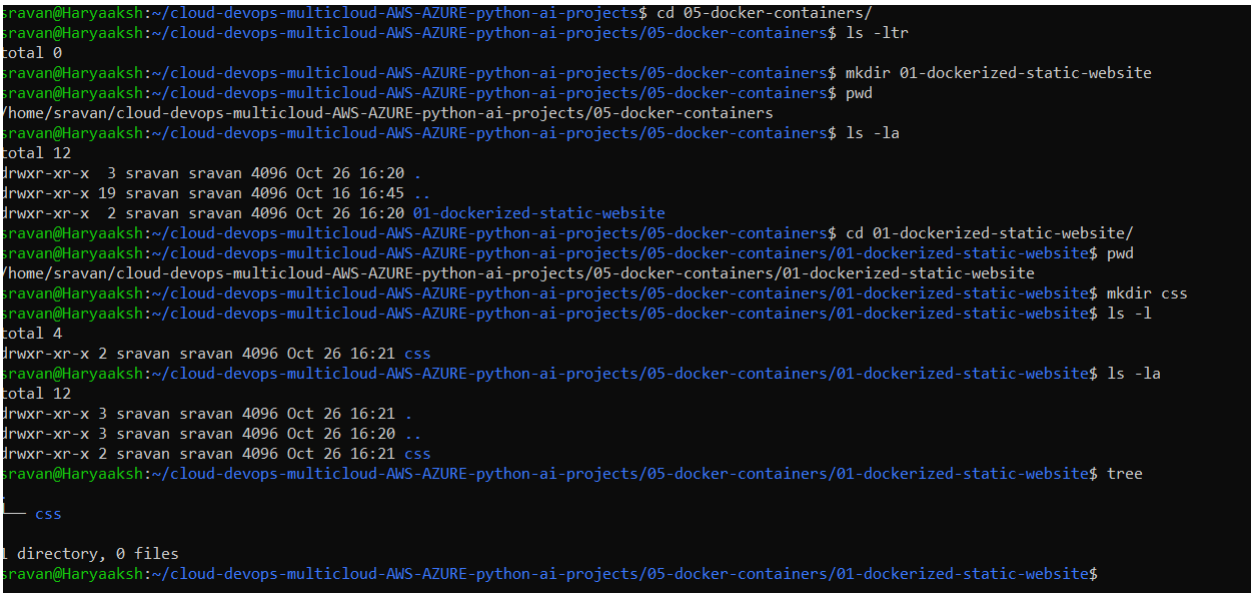
1. Project Overview

This project demonstrates how to containerize and serve a static HTML/CSS website using Docker and Nginx.

It is part of my hands-on Cloud and DevOps learning portfolio. The project covers Docker image creation, container deployment, port mapping, container access, and HTTP verification.

Step 1 — Create the project folder

cd ~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects
cd 05-docker-containers/
mkdir 01-dockerized-static-website
cd 05-docker-containers/01-dockerized-static-website
mkdir css



Step 2 — Create your HTML website

We'll create a small professional-looking Cloud & DevOps portfolio landing page using only HTML and CSS. This also gives a useful screenshot for your GitHub project

vi index.hemls

Step 3 — Create the CSS stylesheet

vi css/style.css

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ vim index.html
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ vi css/style.css
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ ls -la
total 16
drwxr-xr-x 3 sravan sravan 4096 Oct 26 16:29 .
drwxr-xr-x 3 sravan sravan 4096 Oct 26 16:20 ..
drwxr-xr-x 2 sravan sravan 4096 Oct 26 16:30 css
-rw-r--r-- 1 sravan sravan 1826 Oct 26 16:29 index.html
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ ls -R
.:
css  index.html

./css:
style.css
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ docker --version
Docker version 29.8.2, build 7fc2dff
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ docker psa
docker: unknown command: docker psa

Run 'docker --help' for more information
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED       STATUS       PORTS
                          NAMES
fcaf350d47ec   gcr.io/k8s-minikube/kicbase:v0.0.51   "/usr/local/bin/entr…"   3 weeks ago   Up 3 hours   127.0.0.1:52535->22/tcp, 127.0.0.1:52536->237
127.0.0.1:52539->32443/tcp   minikube
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ tree
.
├── css
│   └── style.css
└── index.html

1 directory, 2 files
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$
```

commit the huml and css file

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ ls -ltr
total 8
-rw-r--r-- 1 sravan sravan 1826 Oct 26 16:29 index.html
drwxr-xr-x 2 sravan sravan 4096 Oct 26 16:30 css
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ git add index.html css/style.css
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   css/style.css
        new file:   index.html

sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ git commit -m "feat: add static website HTML and CSS"
[main 6ca720f] feat: add static website HTML and CSS
 2 files changed, 168 insertions(+)
 create mode 100644 05-docker-containers/01-dockerized-static-website/css/style.css
 create mode 100644 05-docker-containers/01-dockerized-static-website/index.html
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ git push origin main
Username for 'https://github.com': sravanboosa
Password for 'https://sravanboosa@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.74 KiB | 1.74 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SravanBoosa/cloud-devops-multicloud-AWS-AZURE-python-ai-projects.git
   7d9ba26..6ca720f  main -> main
```

Step 3— Create the Dockerfile

vi Dockerfile

FROM nginx:alpine

LABEL maintainer="Cloud DevOps Portfolio"

LABEL description="Dockerized static portfolio website"

COPY index.html /usr/share/nginx/html/index.html

COPY css/ /usr/share/nginx/html/css/

EXPOSE 80

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ vi Dockerfile
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ cat  Dockerfile
FROM nginx:alpine

LABEL maintainer="Cloud DevOps Portfolio"
LABEL description="Dockerized static portfolio website"

COPY index.html /usr/share/nginx/html/index.html
COPY css/ /usr/share/nginx/html/css/

EXPOSE 80
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$
```

Instruction	Meaning
FROM nginx:alpine	Uses an Alpine-based Nginx image as the starting point.
LABEL	Adds descriptive metadata to your image.
COPY index.html ...	Copies your HTML page into Nginx's default web root.
COPY css/ ...	Copies the CSS directory into the image.
EXPOSE 80	Documents the container's web-server port. It does not publish that port to your host.

Step 4 — Create .dockerignore

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ vi .dockerignore
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ cat .dockerignore

.git
.gitignore
.env
*.pem
*.key
*.log
.DS_Store
```

Docker excludes these files from the build context, helping avoid sending Git metadata, environment files, private keys, and unnecessary logs to the image build process.

find . -maxdepth 2 -type f

```
sravan@Haryaaksh:~/c
./.dockerignore
./css/style.css
./Dockerfile
./index.html
```

Step 5 — Build the Docker image

docker build -t dockerized-static-website:v1 .

docker images

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ docker images
IMAGE                                                                      ID             DISK USAGE   CONTENT SIZE   EXTRA
dockerized-static-website:v1                                               00c04cd0a8bd       93.7MB         26.3MB
gcr.io/k8s-minikube/kicbase:v0.0.51                                        146cd636030e       1.89GB          532MB
gcr.io/k8s-minikube/kicbase@sha256:4a1c825b61479e6c898851ea66f13c620aaeab6002746e95067fc2c4b38a0b24   4a1c825b6147       1.89GB          532MB    U
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$
```

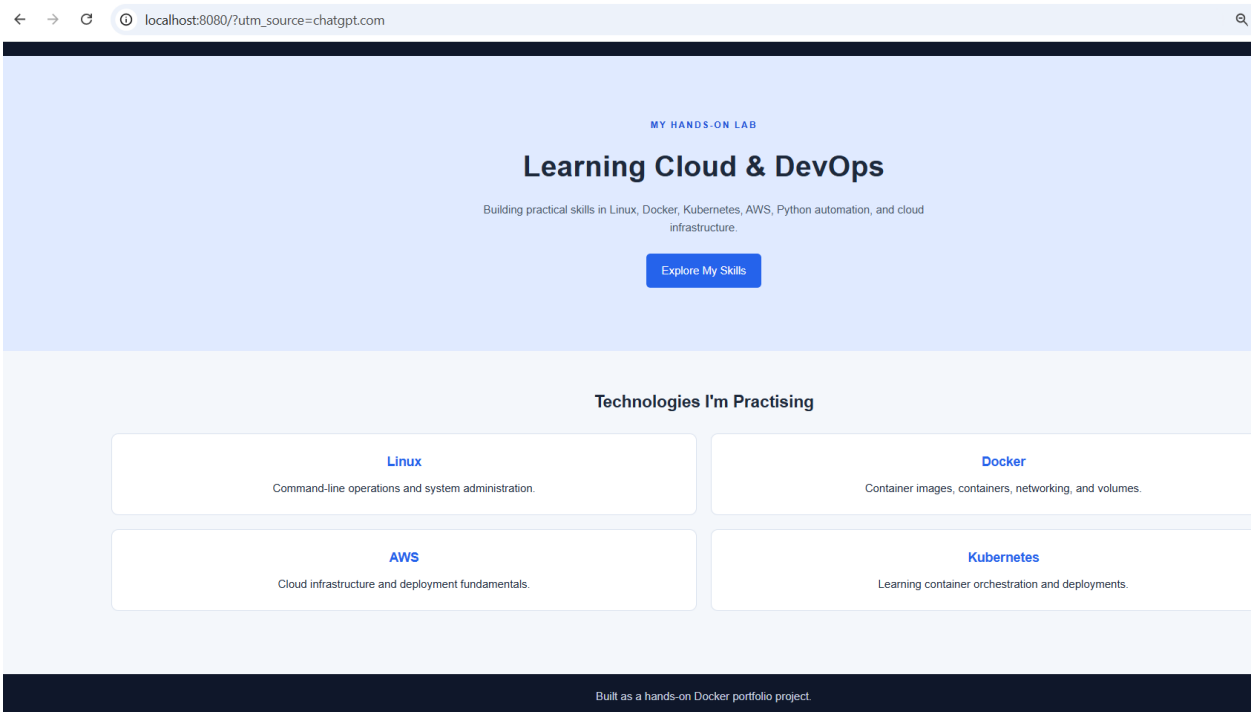
image created successfully

Step 6 — Run the website container

docker run -d --name static-site -p 8080:80 dockerized-static-website:v1

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ docker run -d --name static-site -p 8080:80 dockerized-static-website:v1
d8958169892ef0b94c93756834aa4627e0e2d404fa63e22e977a580cbd1c2a66
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/05-docker-containers/01-dockerized-static-website$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS         PORTS                                                                
```

visit http://localhost:8080/



Enter the container docker exec -it static-site sh

```
/ # curl -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.31.8
Date: Mon, 26 Oct 2026 11:37:51 GMT
Content-Type: text/html
Content-Length: 1826
Last-Modified: Mon, 26 Oct 2026 10:59:23 GMT
Connection: keep-alive
ETag: "6adf328b-722"
Accept-Ranges: bytes

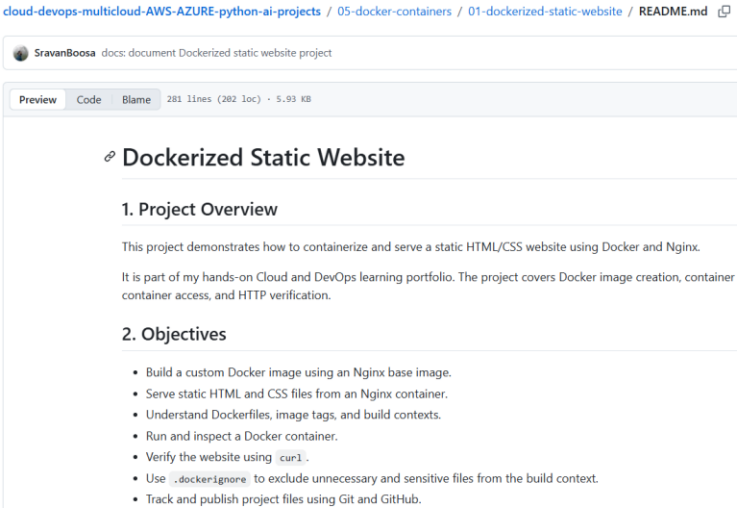
/ # curl http://localhost
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Cloud & DevOps Portfolio</title>
    <link rel="stylesheet" href="css/style.css">
</head>
<body>
    <header>
        <nav class="navbar">
            <h2>CloudPortfolio</h2>
            <span>Docker Learning Project</span>
        </nav>
    </header>

    <main>
        <section class="hero">
            <p class="tag">MY HANDS-ON LAB</p>
            <h1>Learning Cloud & DevOps</h1>
            <p class="description">
                Building practical skills in Linux, Docker, Kubernetes,
                AWS, Python automation, and cloud infrastructure.
            </p>
            <a class="button" href="#skills">Explore My Skills</a>
        </section>
```

Add Readme.md file with all the required information

```
ravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai
rw-r--r-- 1 sravan sravan   51 Oct 26 16:50 .dockerignore
rw-r--r-- 1 sravan sravan  215 Oct 26 16:45 Dockerfile
rw-r--r-- 1 sravan sravan 6069 Oct 26 17:18 README.md
rw-r--r-- 1 sravan sravan 1570 Oct 26 16:30 css/style.css
rw-r--r-- 1 sravan sravan 1826 Oct 26 16:29 index.html
```

Stage and commit the file Dockerfile .dockerignore README.md→Push to Github



docker stop static-site